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Did US Politicians Expect the China Shock?

Contents

1	Big picture	2
2	Notation	2
3	Model 1: vote choice with expected future China shock	2
3.1	Electoral support and accountability	2
3.2	Legislator utility and the choice rule	3
3.3	Vote probability (probit)	3
4	Model 2: MLE when the information set is fully specified	3
4.1	The three information-set scenarios	4
5	Model 3: moment-inequality approach when information is only partially specified	4
5.1	Odds-based moment inequalities	4
5.2	Revealed-preference moment inequalities	4
6	Construction of the China shock variables (the empirical “state” S)	5
6.1	Commuting-zone level: future and past shocks	5
6.2	Aggregation to congressional districts	5
7	Figures	5
8	Tables	7
9	A short summary	14
10	Conclusion	14
10.1	One sentence	14
10.2	What is the paper doing?	15
10.3	Core intuition: why expectations matter (the key identification problem)	15
10.4	Main findings	15
10.5	So what is the conclusion, conceptually?	16
10.6	Why this is useful	16

1 Big picture

- Setting: repeated U.S. House roll call votes about renewing China’s Normal Trade Relations (NTR) status, 1990–2001.
- Question: did legislators *anticipate* the future local labor-market exposure to import competition from China when they voted on trade policy?
- Strategy: model the vote as a probit choice where the key state is the legislator’s *expected future* “China shock” in the district. Then test which information set (minimal, baseline, perfect foresight) is consistent with observed votes using (i) standard MLE when the information set is fully specified and (ii) moment-inequality inference when it is only partially specified.
- Main message I take away: legislators appear to have had *substantial* information about future China-related import pressure (more than “minimal”), but *less than perfect foresight*, and information/predictability declines over time. Ideology matters more than constituency trade exposure, but constituency exposure matters with the expected sign.

2 Notation

- i indexes legislators (House members), t indexes roll call votes (yearly in the main figures/tables).
- $Y_{i,t} \in \{0, 1\}$ indicates a pro-China vote (vote to renew/extend/protect China’s NTR status).
- Two policy “locations” on a one-dimensional policy line:
 - x_t : the pro-China policy alternative (renew NTR).
 - q_t : the anti-China policy alternative (do not renew / restrict).
- θ_i : legislator ideology (time-invariant in the baseline setup).
- $S_{i,t+1}$: district exposure to the future China import shock relevant for the vote at t .
- $\mathcal{I}_{i,t}$: the information set of legislator i at the time of the vote.
- $\xi_{i,t,x}$ and $\xi_{i,t,q}$: idiosyncratic taste shocks for each alternative; define $\xi_{i,t} \equiv \xi_{i,t,q} - \xi_{i,t,x} \sim \mathcal{N}(0, 1)$ (normalization delivers a probit).

3 Model 1: vote choice with expected future China shock

3.1 Electoral support and accountability

The paper writes next-period electoral support as

$$V_{i,t+1} = h_t(d_{i,t}, S_{i,t+1}) + e_{i,t+1}, \quad (1)$$

$$\mathbb{E}[e_{i,t+1} \mid d_{i,t}, S_{i,t+1}, \mathcal{I}_{i,t}] = 0, \quad (2)$$

where $d_{i,t} \in \{x_t, q_t\}$ is the policy choice (vote) and $S_{i,t+1}$ is the future shock.

A simple linear-in- S form for h_t is

$$h_t(d_{i,t}, S_{i,t+1}) = \gamma_t^0 + \gamma_t^2 S_{i,t+1} + (\gamma_t^1 - \gamma_t^2) S_{i,t+1} \mathbf{1}\{d_{i,t} = x_t\}. \quad (3)$$

interpretation: the difference in expected electoral support between choosing x_t versus q_t is proportional to the expected future shock:

$$\mathbb{E}[V_{i,t+1} \mid d_{i,t} = x_t, \mathcal{I}_{i,t}] - \mathbb{E}[V_{i,t+1} \mid d_{i,t} = q_t, \mathcal{I}_{i,t}] = (\gamma_t^1 - \gamma_t^2) \mathbb{E}[S_{i,t+1} \mid \mathcal{I}_{i,t}].$$

3.2 Legislator utility and the choice rule

Utility from choosing $d_{i,t} \in \{x_t, q_t\}$:

$$U(\xi_{i,t}, d_{i,t}; \theta_i, \mathcal{I}_{i,t}) = u(\|d_{i,t} - \theta_i\|) + \tilde{\delta} \mathbb{E}[V_{i,t+1} \mid d_{i,t}, \mathcal{I}_{i,t}] + \xi_{i,t,d}. \quad (4)$$

Let $Y_{i,t} = 1$ denote voting for x_t (pro-China). Under the quadratic-distance form used in the paper, the latent index for $Y_{i,t}$ is

$$Y_{i,t} = \mathbf{1} \left\{ -\frac{1}{2} \left[(x_t - \theta_i)^2 - (q_t - \theta_i)^2 \right] + \tilde{\delta} \left(\mathbb{E}[V_{i,t+1} \mid x_t, \mathcal{I}_{i,t}] - \mathbb{E}[V_{i,t+1} \mid q_t, \mathcal{I}_{i,t}] \right) \geq \xi_{i,t} \right\}. \quad (5)$$

Plugging in the linear $h_t(\cdot)$ representation, define the key reduced-form parameters

$$\delta_t \equiv \tilde{\delta}(\gamma_t^1 - \gamma_t^2), \quad a_t \equiv x_t - q_t, \quad b_t \equiv \frac{1}{2}(q_t^2 - x_t^2). \quad (6)$$

Then the choice rule becomes

$$Y_{i,t} = \mathbf{1} \{ a_t \theta_i + b_t + \delta_t \mathbb{E}[S_{i,t+1} \mid \mathcal{I}_{i,t}] \geq \xi_{i,t} \}. \quad (7)$$

3.3 Vote probability (probit)

Given $\xi_{i,t} \sim \mathcal{N}(0, 1)$,

$$\Pr(Y_{i,t} = 1 \mid \mathcal{I}_{i,t}) = \Phi(a_t \theta_i + b_t + \delta_t \mathbb{E}[S_{i,t+1} \mid \mathcal{I}_{i,t}]). \quad (8)$$

How I read parameters:

- a_t loads ideology into the latent index (a “slope” on θ_i).
- b_t is a period-specific intercept (baseline propensity to vote pro-China holding ideology and expected shocks fixed).
- δ_t is the weight on expected future China shock exposure in the district. If $\delta_t < 0$, then anticipating a larger future shock reduces the pro-China vote probability.

4 Model 2: MLE when the information set is fully specified

If I posit an information set (so $\mathbb{E}[S_{i,t+1} \mid \mathcal{I}_{i,t}]$ is pinned down), the log-likelihood for a given period is the standard probit likelihood:

$$\log L(\omega_t) = \sum_i Y_{i,t} \log \Phi(z_{i,t}(\omega_t)) + \sum_i (1 - Y_{i,t}) \log(1 - \Phi(z_{i,t}(\omega_t))), \quad (9)$$

where $\omega_t = (a_t, b_t, \delta_t)$ and

$$z_{i,t}(\omega_t) = a_t \theta_i + b_t + \delta_t \mathbb{E}[S_{i,t+1} \mid \mathcal{I}_{i,t}].$$

4.1 The three information-set scenarios

The paper operationalizes three nested “knowledge” cases (paraphrase):

- **Minimal information:** legislators observe their ideology and a coarse district characteristic (manufacturing employment share), but not the current shock. Implemented by forecasting $S_{i,t+1}$ via

$$S_{i,t+1} = \beta_0 + \beta_1 \theta_i + \beta_2 \text{ShareMfg}_{i,t} + \varepsilon_{i,t+1},$$

and using $\widehat{\mathbb{E}}[S_{i,t+1} \mid \mathcal{I}_{i,t}]$ from this regression.

- **Baseline information:** legislators also observe the *current* district shock $S_{i,t}$ (plus ideology and manufacturing share). Implemented via

$$S_{i,t+1} = \beta_0 + \beta_1 \theta_i + \beta_2 \text{ShareMfg}_{i,t} + \beta_3 S_{i,t} + \varepsilon_{i,t+1}.$$

- **Perfect foresight:** legislators know $S_{i,t+1}$ itself (or, equivalently for testing, they know $S_{i,t+1}$ plus an additional orthogonal component beyond the baseline predictors).

5 Model 3: moment-inequality approach when information is only partially specified

The key issue is that I do not observe $\mathcal{I}_{i,t}$. The paper proposes testing candidate information sets using conditional moment inequalities.

Let $Z_{i,t} \subseteq \mathcal{I}_{i,t}$ be observable covariates that are assumed to be in the true information set. In applications, $Z_{i,t}$ corresponds to the minimal or baseline set (e.g., $\{\text{ShareMfg}_{i,t}, \theta_i\}$ or $\{S_{i,t}, \text{ShareMfg}_{i,t}, \theta_i\}$).

5.1 Odds-based moment inequalities

Starting from the choice rule and using properties of the probit model, the paper constructs two conditional moments:

$$m_{ob}^\ell(\omega_t) \equiv (1 - Y_{i,t}) \frac{\Phi(a_t \theta_i + b_t + \delta_t S_{i,t+1})}{1 - \Phi(a_t \theta_i + b_t + \delta_t S_{i,t+1})} - Y_{i,t}, \quad (10)$$

$$m_{ob}^u(\omega_t) \equiv Y_{i,t} \frac{1 - \Phi(a_t \theta_i + b_t + \delta_t S_{i,t+1})}{\Phi(a_t \theta_i + b_t + \delta_t S_{i,t+1})} - (1 - Y_{i,t}). \quad (11)$$

Then, at the true parameter ω_t^0 ,

$$\mathbb{E}[m_{ob}^\ell(\omega_t^0) \mid Z_{i,t}] \geq 0, \quad \mathbb{E}[m_{ob}^u(\omega_t^0) \mid Z_{i,t}] \geq 0. \quad (12)$$

5.2 Revealed-preference moment inequalities

A complementary pair of inequalities is based on revealed preference with the latent index:

$$m_{rp}^\ell(\omega_t) \equiv Y_{i,t} (a_t \theta_i + b_t + \delta_t S_{i,t+1}) + (1 - Y_{i,t}) \frac{\phi(a_t \theta_i + b_t + \delta_t S_{i,t+1})}{1 - \Phi(a_t \theta_i + b_t + \delta_t S_{i,t+1})}, \quad (13)$$

$$m_{rp}^u(\omega_t) \equiv -(1 - Y_{i,t}) (a_t \theta_i + b_t + \delta_t S_{i,t+1}) + Y_{i,t} \frac{\phi(a_t \theta_i + b_t + \delta_t S_{i,t+1})}{\Phi(a_t \theta_i + b_t + \delta_t S_{i,t+1})}. \quad (14)$$

At ω_t^0 ,

$$\mathbb{E}[m_{rp}^\ell(\omega_t^0) \mid Z_{i,t}] \geq 0, \quad \mathbb{E}[m_{rp}^u(\omega_t^0) \mid Z_{i,t}] \geq 0. \quad (15)$$

How I interpret the logic: if I guess a candidate information set $Z_{i,t}$ that is too *small* (i.e., it omits information legislators really used), the implied inequalities may still hold (weak identification). If I guess an information set that is too *large* in the “perfect foresight” sense (it effectively assumes legislators knew the unpredictable component), the inequalities can fail, producing an *empty* confidence set and rejection.

6 Construction of the China shock variables (the empirical “state” S)

The paper builds district exposure to import-supply shocks from China using an industry-by-location shift-share design.

6.1 Commuting-zone level: future and past shocks

For commuting zone j , the *future* shock used for the vote at time t is

$$S_{j,t+1} = \sum_k \frac{L_{j,k,t}}{L_{j,t}} \frac{\Delta M_{k,t+1}^{oth}}{Y_{k,t} + M_{k,t} - X_{k,t}}, \quad (16)$$

where (wording) $\Delta M_{k,t+1}^{oth}$ is the future change in Chinese imports into a set of other high-income countries for industry k , and the denominator normalizes by absorption in the U.S. for industry k at time t .

The corresponding *past* shock is

$$S_{j,t} = \sum_k \frac{L_{j,k,t-5}}{L_{j,t-5}} \frac{\Delta M_{k,t}^{oth}}{Y_{k,t-5} + M_{k,t-5} - X_{k,t-5}}. \quad (17)$$

6.2 Aggregation to congressional districts

The commuting-zone shocks are aggregated to congressional districts using population weights (CZs can overlap district boundaries).

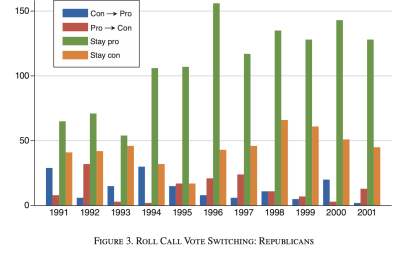
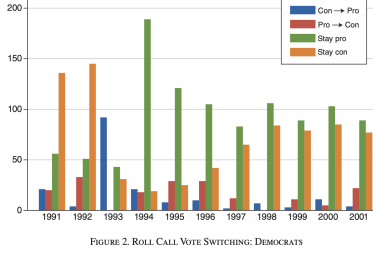
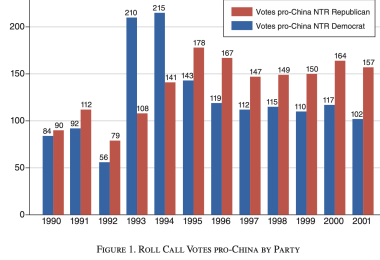
7 Figures

Figure 1: pro-China NTR votes by party (1990–2001)

- What is plotted: counts of pro-China votes in each year, split by party.
- The raw counts (as labeled in the figure) are:

Year	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001
Dem. pro	84	92	56	210	215	143	119	112	115	110	117	102
Rep. pro	90	112	79	108	141	178	167	147	149	150	164	157

- Interpretation: early 1990s show relatively more Democratic support (especially 1993–1994), while later years show strong Republican support (1995 onward). The “China” vote is not stable along a single party line throughout the decade.



Figures 2–3: within-legislator switching, by party

- What is plotted: for each year (from 1991 onward), counts of legislators who (i) switch from con to pro, (ii) switch from pro to con, (iii) stay pro, (iv) stay con, comparing vote t to vote $t - 1$.
- Interpretations: there is meaningful within-legislator variation (not just party composition changing). The paper summarizes the switching rates as roughly on the order of 15–20% per year, which is important for identification because it implies votes respond to time-varying incentives/states.

Figure 4: import shocks and the pro-China vote share over time

- What is plotted: (i) bar = share voting pro-China in each year; (ii) blue line = average *past* district shock $S_{i,t}$; (iii) red line = average *future* district shock $S_{i,t+1}$.
- Interpretations: vote shares peak in the mid-1990s while the shock series evolve over time. The gap between the future shock (what should matter in the forward-looking model) and the past shock (what a backward-looking proxy would use) is nontrivial, which motivates the “expectations” focus.

Figure 5: predictability of the China shock at the industry level

The figure reports autocorrelation estimates α_t from a regression of future industry-level import growth on past import growth (with year fixed effects), i.e.,

$$\Delta \ln M_{k,t,t+5}^{oth} = \alpha_t \Delta \ln M_{k,t-5,t}^{oth} + D_t + \varepsilon_{k,t}. \quad (18)$$

Reading: α_t is around 0.2–0.3 early on but declines toward zero (even slightly negative in the late 1990s). This lines up with the paper’s evidence that perfect foresight is not plausible and that forecasting got harder over time.

Figure 6: why “more information” does not mechanically raise or lower pro-China votes

- The figure is a conceptual diagram: because $\Phi(\cdot)$ is S-shaped, adding noise to the latent index (i.e., averaging over uncertainty) can increase or decrease the expected vote probability depending on whether the baseline index is in the convex or concave region.

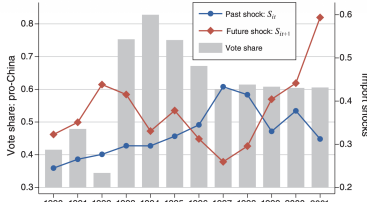


FIGURE 4. IMPORT SHOCKS FROM CHINA AND PRO-CHINA NTR VOTE SHARE

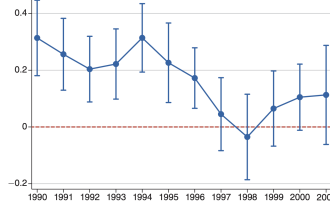


FIGURE 5. THE CHINA SHOCK OVER TIME

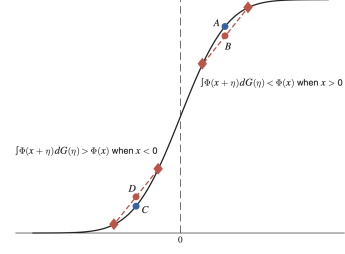


FIGURE 6. AN ILLUSTRATIVE EXAMPLE: EFFECTS OF IMPROVING INFORMATION

- Takeaways: the effect of giving legislators perfect information is theoretically ambiguous; it depends on where each legislator sits on the probit curve.

8 Tables

Table 1: MLE estimates under alternative information assumptions

Table 1: MLE estimates of $(\hat{a}, \hat{b}, \hat{\delta})$ under different information-set assumptions

	$\hat{a}$ (SE)	$\hat{b}$ (SE)	$\hat{\delta}$ (SE)
<i>Panel A: 1997–2001</i>			
Minimal info ($\{\text{ShareMfg}_{i,t}, \theta_i\}$)	0.650 (0.063)	0.262 (0.025)	−0.810 (0.314)
Baseline info ($\{S_{i,t}, \text{ShareMfg}_{i,t}, \theta_i\}$)	0.643 (0.062)	0.263 (0.025)	−0.786 (0.279)
Perfect foresight ($S_{i,t+1}$ known)	0.619 (0.062)	0.263 (0.025)	−0.018 (0.211)
<i>Panel B: 1993–1996</i>			
Minimal info ($\{\text{ShareMfg}_{i,t}, \theta_i\}$)	−0.046 (0.084)	0.678 (0.033)	−0.440 (0.611)
Baseline info ($\{S_{i,t}, \text{ShareMfg}_{i,t}, \theta_i\}$)	−0.043 (0.084)	0.679 (0.033)	−1.116 (0.549)
Perfect foresight ($S_{i,t+1}$ known)	−0.051 (0.083)	0.678 (0.033)	−0.517 (0.452)
<i>Panel C: 1990–1992</i>			
Minimal info ($\{\text{ShareMfg}_{i,t}, \theta_i\}$)	1.019 (0.104)	−0.179 (0.037)	0.065 (0.351)
Baseline info ($\{S_{i,t}, \text{ShareMfg}_{i,t}, \theta_i\}$)	1.000 (0.105)	−0.181 (0.037)	−0.981 (0.288)
Perfect foresight ($S_{i,t+1}$ known)	1.013 (0.104)	−0.180 (0.037)	−0.669 (0.244)
<i>Panel D: pooled 1990–2001</i>			
Minimal info ($\{\text{ShareMfg}_{i,t}, \theta_i\}$)	0.532 (0.044)	0.280 (0.017)	−0.453 (0.222)
Baseline info ($\{S_{i,t}, \text{ShareMfg}_{i,t}, \theta_i\}$)	0.534 (0.044)	0.280 (0.017)	−0.856 (0.198)
Perfect foresight ($S_{i,t+1}$ known)	0.523 (0.044)	0.280 (0.017)	−0.333 (0.147)

note: “Minimal” and “baseline” implement $\mathbb{E}[S_{i,t+1} \mid \mathcal{I}_{i,t}]$ using an OLS prediction; “perfect foresight” plugs in the realized $S_{i,t+1}$. Standard errors are in parentheses.

Interpretation (Table 1).

- The constituency term $\hat{\delta}$ is negative in most cases except when the information assumption is very weak (minimal info in 1990–1992 yields a small positive estimate). Once I allow baseline information, $\hat{\delta}$ is negative throughout, consistent with “more expected China import exposure $\Rightarrow$ less pro-China voting.”
- The perfect-foresight assumption can *attenuate* $\hat{\delta}$ strongly, especially in 1997–2001 where it becomes near zero. read is that if I force the model to use realized $S_{i,t+1}$ when legislators did not truly observe it, the coefficient on $S_{i,t+1}$ gets pushed toward zero.

- $\hat{a}$ and $\hat{b}$ are fairly stable across information assumptions (relative to $\hat{\delta}$). This matches the idea that ideology and baseline pro-trade propensity are easier to infer from votes than the subtle expectations term.

Table 2: how predictable are future shocks from baseline covariates?

Table 2: Predicting future district shocks with past shocks and manufacturing share (OLS)

	1990–2001 (1)	1997–2001 (2)	1993–1996 (3)	1990–1992 (4)
$S_{i,t}$	0.404 (0.017)	0.731 (0.041)	0.214 (0.013)	0.654 (0.033)
$ShareMfg_{i,t}$	0.705 (0.021)	0.680 (0.036)	0.457 (0.020)	0.700 (0.045)
Observations	5,494	2,564	1,698	1,232
R^2	0.556	0.584	0.674	0.695

note: Dependent variable is $S_{i,t+1}$. Standard errors in parentheses.

Interpretation (Table 2).

- Baseline observables ($S_{i,t}$ and manufacturing share) explain a lot of the variation in $S_{i,t+1}$: R^2 is about 0.56 pooled and as high as 0.67–0.70 in earlier subsamples.
- The coefficient on lagged shock $S_{i,t}$ is large in 1997–2001 (0.731), suggesting persistence where it exists; but the paper’s Figure 5 suggests that industry-level persistence declines later, so predictability is not constant over time.
- Manufacturing share is always strongly predictive of future shocks (loadings around 0.46–0.71), consistent with a shift-share structure where exposure is mechanically related to industrial composition.

Table 3: moment-inequality confidence sets and specification testsTable 3: 95% confidence sets for (a, b, δ) and p-values from information-set specification tests

Period	Info assumption	CS of a	CS of b	CS of δ	p (BP)	p (RC)	p (RS)	N
<i>1997–2001</i>								
	Minimal ($\{\text{ShareMfg}, \theta\}$)	[0.450, 0.720]	[0.165, 0.260]	[−1.975, −0.225]	0.405	0.405	0.405	2,564
	Baseline ($\{S_t, \text{ShareMfg}, \theta\}$)	[0.435, 0.750]	[0.185, 0.290]	[−1.560, −0.075]	0.290	0.275	0.275	2,564
	Perfect foresight (too strong)	—	—	—	0.010	0.010	0.010	2,564
<i>1993–1996</i>								
	Minimal ($\{\text{ShareMfg}, \theta\}$)	[−0.280, 0.100]	[0.583, 0.703]	[−2.300, 0.900]	0.330	0.330	0.330	1,698
	Baseline ($\{S_t, \text{ShareMfg}, \theta\}$)	[−0.325, 0.130]	[0.598, 0.740]	[−3.125, −0.125]	0.395	0.395	0.395	1,698
	Perfect foresight (too strong)	—	—	—	0.040	0.035	0.035	1,698
<i>1990–1992</i>								
	Minimal ($\{\text{ShareMfg}, \theta\}$)	[0.800, 1.550]	[−0.325, −0.125]	[−1.125, 2.125]	0.955	0.955	0.955	1,232
	Baseline ($\{S_t, \text{ShareMfg}, \theta\}$)	[1.025, 1.438]	[−0.275, −0.150]	[−1.300, 0.000]	0.165	0.145	0.145	1,232
	Perfect foresight	[1.000, 1.550]	[−0.270, −0.128]	[−1.624, 0.096]	0.255	0.240	0.240	1,232
<i>1990–2001 (pooled)</i>								
	Minimal ($\{\text{ShareMfg}, \theta\}$)	[0.463, 0.625]	[0.210, 0.270]	[−1.163, −0.150]	0.185	0.185	0.185	5,494
	Baseline ($\{S_t, \text{ShareMfg}, \theta\}$)	[0.463, 0.650]	[0.235, 0.287]	[−1.438, −0.500]	0.190	0.190	0.190	5,494
	Perfect foresight (too strong)	—	—	—	0.010	0.010	0.010	5,494

note: “—” means the 95% confidence set is empty, which the authors interpret as rejection of that information-set hypothesis. BP/RC/RS are three different test statistics reported in the paper.

Interpretation (Table 3).

- Across the full sample and especially post-1993, δ is tightly negative under baseline information. That is the core evidence that expected China exposure reduced support for renewing NTR.
- “Perfect foresight” is rejected in 1993–1996, 1997–2001, and pooled 1990–2001 (empty sets with small p-values). In 1990–1992 perfect foresight is not rejected, which I interpret as: early shocks may have been more predictable, so assuming near-perfect knowledge is less crazy early on.
- Minimal information delivers weaker identification of δ (e.g., it includes positive values in 1993–1996 and 1990–1992). Takeaways: leaving out $S_{i,t}$ makes expectations too noisy and the model struggles to pin down the direction of the constituency effect.
- The ideology loading a varies by subperiod (even straddling zero in 1993–1996), which I read as: the ideological meaning of the NTR vote changes across political eras.

Table 4: heterogeneity by party (baseline information)

Table 4: Party heterogeneity: 95% confidence sets and information-set tests (baseline information)

Period	Group	CS of a	CS of b	CS of δ	p (BP)	p (RC)	p (RS)	N
<i>1997–2001</i>								
	Democrats	[2.225, 4.625]	[0.850, 1.800]	[−4.425, −0.015]	0.315	0.265	0.265	1,229
	Republicans	–	–	–	0.010	0.010	0.010	1,326
<i>1993–1996</i>								
	Democrats	[0.800, 3.400]	[1.050, 2.050]	[−6.325, −0.025]	0.380	0.380	0.380	888
	Republicans	–	–	–	0.020	0.015	0.015	806
<i>1990–1992</i>								
	Democrats	[1.875, 4.375]	[0.050, 0.850]	[−3.700, −0.400]	0.415	0.410	0.410	745
	Republicans	[−1.300, 1.400]	[−0.300, 0.650]	[−1.500, 1.800]	0.315	0.310	0.310	485
<i>1990–2001 (pooled)</i>								
	Democrats	[1.375, 3.325]	[0.570, 1.290]	[−4.412, −0.417]	0.990	0.990	0.990	2,862
	Republicans	–	–	–	0.010	0.010	0.010	2,617

note: Baseline information means $Z_{i,t} = \{S_{i,t}, ShareMfg_{i,t}, \theta_i\}$ is assumed observed by legislators. “–” indicates rejection/empty set.

Interpretation (Table 4).

- For Democrats, δ is negative and the baseline information assumption is not rejected in every subperiod. So (i) Democrats look “informed enough” under baseline and (ii) they appear to weigh the constituency exposure negatively when voting pro-China.
- For Republicans, the baseline information assumption is rejected in 1993–1996 and 1997–2001 (and pooled). This is striking: it suggests either (a) Republicans used a different information set / different mapping from observables to expectations, or (b) the model fit is worse for Republicans under the baseline restriction.
- The ideology parameter a is much larger for Democrats (especially after 1993). read: the latent ideological “direction” of the vote is party-specific, so splitting by party changes the scale/meaning of θ in the probit index.

Table 5: heterogeneity by electoral pressure (win margin)

Table 5: Heterogeneity by prior win margin (baseline information)

Period	Group	CS of a	CS of b	CS of δ	p (BP)	p (RC)	p (RS)	N
<i>1997–2001</i>								
	Win margin > median	[0.600, 1.225]	[0.060, 0.270]	[−1.375, 1.500]	0.890	0.890	0.890	1,205
	Win margin < median	[−0.125, 0.500]	[0.250, 0.438]	[−3.413, −0.525]	0.680	0.675	0.675	1,207
<i>1993–1996</i>								
	Win margin > median	[−0.375, 0.350]	[0.475, 0.700]	[−3.250, 2.750]	0.560	0.560	0.560	799
	Win margin < median	[−1.050, 0.300]	[0.625, 0.975]	[−8.550, −0.225]	0.965	0.965	0.965	800
<i>1990–1992</i>								
	Win margin > median	[0.875, 2.125]	[−0.465, −0.105]	[−2.875, 3.312]	0.765	0.755	0.755	568
	Win margin < median	[0.450, 1.700]	[−0.325, −0.025]	[−4.375, −0.750]	0.510	0.465	0.465	569
<i>1990–2001 (pooled)</i>								
	Win margin > median	[0.550, 0.888]	[0.120, 0.232]	[−0.900, 0.800]	0.450	0.450	0.450	2,573
	Win margin < median	[0.013, 0.475]	[0.282, 0.410]	[−2.565, −0.825]	0.615	0.615	0.615	2,575

note: Baseline information assumed. “Win margin” refers to the previous election.

Interpretation (Table 5).

- Tight races (win margin below median) show δ that is entirely negative across all subsamples, while safe seats often have δ intervals that include zero or even positive values. This matches the idea that electorally vulnerable legislators are more responsive to constituency-specific labor-market stakes.
- I do *not* see a consistent pattern that tight races have systematically “more information” (p-values are not systematically smaller), so the difference looks like preferences/weights rather than forecasting ability.

Table 6: heterogeneity by tenure

Table 6: Heterogeneity by tenure (baseline information)

Period	Group	CS of a	CS of b	CS of δ	p (BP)	p (RC)	p (RS)	N
<i>1997–2001</i>								
	Tenure > median	[0.750, 1.075]	[0.158, 0.255]	[−0.850, 0.600]	0.120	0.105	0.105	1,372
	Tenure < median	[0.180, 0.700]	[0.238, 0.375]	[−2.750, −0.875]	0.185	0.165	0.165	1,176
<i>1993–1996</i>								
	Tenure > median	[−0.400, 0.350]	[0.525, 0.700]	[−3.350, 1.363]	0.460	0.460	0.460	863
	Tenure < median	[−0.700, 0.237]	[0.650, 0.925]	[−5.875, 0.312]	0.630	0.615	0.615	821
<i>1990–1992</i>								
	Tenure > median	[0.825, 2.400]	[−0.175, 0.175]	[−5.513, 0.175]	0.920	0.920	0.920	572
	Tenure < median	[0.938, 1.750]	[−0.495, −0.270]	[−1.600, 0.900]	0.260	0.235	0.235	634
<i>1990–2001 (pooled)</i>								
	Tenure > median	[0.615, 0.965]	[0.200, 0.287]	[−1.758, −0.270]	0.275	0.275	0.275	2,488
	Tenure < median	[0.238, 0.475]	[0.262, 0.338]	[−1.502, −0.355]	0.145	0.135	0.135	2,950

note: Tenure split is relative to the sample median within the relevant period.

Interpretation (Table 6).

- Junior legislators (tenure below median) show more negative δ in 1997–2001, while senior legislators’ δ includes zero. This could reflect electoral incentives (newer members may be more sensitive), or selection (different types).
- However, across periods the differences are not clean. I agree with the paper’s tone: there is not strong evidence that experience systematically changes either information or preferences in this setting.

Table 7: heterogeneity by business contributions

Table 7: Heterogeneity by campaign contributions from business (baseline information)

Period	Group	CS of a	CS of b	CS of δ	p (BP)	p (RC)	p (RS)	N
<i>1997–2001</i>								
	Money > median	[−0.350, 0.450]	[0.492, 0.720]	[−2.850, −0.600]	0.360	0.360	0.360	1,273
	Money < median	[0.525, 1.078]	[−0.045, 0.135]	[−2.200, 0.425]	0.380	0.345	0.345	1,273
<i>1993–1996</i>								
	Money > median	[−1.000, 0.250]	[0.725, 1.075]	[−7.000, −0.400]	0.715	0.695	0.695	844
	Money < median	[−0.312, 0.588]	[0.450, 0.700]	[−3.650, 2.125]	0.665	0.665	0.665	844
<i>1990–1992</i>								
	Money > median	[0.150, 1.100]	[−0.185, 0.025]	[−2.038, 0.325]	0.370	0.370	0.370	613
	Money < median	[1.112, 3.475]	[−0.400, 0.275]	[−7.325, 0.775]	0.830	0.830	0.830	616
<i>1990–2001 (pooled)</i>								
	Money > median	[0.052, 0.438]	[0.390, 0.490]	[−2.090, −0.900]	0.220	0.215	0.215	2,730
	Money < median	[0.510, 0.912]	[0.100, 0.220]	[−1.770, 0.128]	0.440	0.440	0.440	2,733

note: Money split is based on business contributions being above/below the median.

Interpretation (Table 7).

- Differences by business money exist but are not dramatic: confidence sets overlap a lot.
- One pattern I notice: in later periods (1993–1996, 1997–2001, pooled), the high-business-money group tends to have δ that is more clearly negative, while the low-money group sometimes includes zero. This could be consistent with business donors being attentive to trade shocks, but the evidence is not sharp.

Table 8: placebo/validation using NTR votes for Vietnam

Table 8: Validation exercise: NTR votes for Vietnam (1998–2001)

Info assumption	CS of a	CS of b	CS of δ	p (BP)	p (RC)	p (RS)	N
Minimal ($\{\text{ShareMfg}, \theta\}$)	[−1.300, −0.300]	[0.600, 0.938]	[−85.000, −21.625]	0.990	0.990	0.990	1,595
Baseline ($\{S_t^V, \text{ShareMfg}, \theta\}$)	[−1.250, −0.613]	[0.630, 0.792]	[−74.075, −6.700]	0.435	0.435	0.435	1,595
Perfect foresight (too strong)	—	—	—	0.025	0.020	0.020	1,595

note: S_t^V denotes the Vietnam-specific shock measure used in this validation setting.

Interpretation (Table 8).

- The “perfect foresight” hypothesis is again rejected, which strengthens the Interpretation that the method is genuinely detecting unrealistic information assumptions rather than mechanically failing in the China-only application.
- The δ magnitudes are huge in absolute value here. interpretation: the Vietnam NTR “shock” variation is on a different scale than the China shock (or the policy stakes differ), so δ is not directly comparable across contexts without rescaling.

Counterfactual simulation objects (the “model” behind Tables 9–10)

Given parameters ω_t and an information set $\mathcal{I}_{i,t}$, the model-implied pro-China probability is

$$p_{i,t}(\omega_t, \mathcal{I}_{i,t}) = \Pr(Y_{i,t} = 1 \mid \mathcal{I}_{i,t}) = \Phi(a_t \theta_i + b_t + \delta_t \mathbb{E}[S_{i,t+1} \mid \mathcal{I}_{i,t}]) .$$

Define the model-implied pro-China vote share in a period with N_t legislators as

$$\Pi^+(\omega_t, \mathcal{I}_t, N_t) = \frac{1}{N_t} \sum_{i=1}^{N_t} p_{i,t}(\omega_t, \mathcal{I}_{i,t}).$$

The “effect of improving information” counterfactual compares baseline information $\mathcal{I}_{i,t}^b$ to perfect information $\mathcal{I}_{i,t}^p$. Because ω_t is set-identified, the paper reports a range (paraphrase of their equation (20)):

$$\left[\min_{\omega_t \in \Omega_t^{0.95}} \left(\Pi^+(\omega_t, \mathcal{I}_t^p, N_t) - \Pi^+(\omega_t, \mathcal{I}_t^b, N_t) \right) \times 100, \max_{\omega_t \in \Omega_t^{0.95}} \left(\Pi^+(\omega_t, \mathcal{I}_t^p, N_t) - \Pi^+(\omega_t, \mathcal{I}_t^b, N_t) \right) \times 100 \right], \quad (19)$$

where $\Omega_t^{0.95}$ is the 95% confidence region for ω_t .

Table 9: counterfactual — giving legislators perfect information

Table 9: Counterfactual: changing the information set from baseline to perfect information

	1997–2001	1993–1996	1990–1992
1. Change in pro-CHN vote share (%)	[−0.030, 0.012]	[−0.161, −0.000]	[−0.008, 0.064]
2. Always pro-CHN (%)	[55.945, 61.671]	[70.389, 76.967]	[36.626, 42.357]
3. Switch pro → against (%)	[0.077, 1.701]	[0.054, 1.618]	[0.000, 1.262]
4. Switch against → pro (%)	[0.078, 1.694]	[0.054, 1.486]	[0.000, 1.308]
5. Always against-CHN (%)	[36.736, 42.196]	[21.985, 27.637]	[56.268, 62.314]

note: The table reports ranges coming from uncertainty in the estimated parameters. “Baseline” is assumed to be the true information set in this simulation.

Interpretation (Table 9).

- The counterfactual change in the aggregate pro-China vote share is tiny in all periods (close to zero percentage points). This is a key punchline: *lack of information* is not what made the bills pass.
- Even when information changes, the share of actual switchers is small: only about 0–1.7% switch in either direction depending on the period.
- The sign of the change can be slightly negative or positive depending on the period, consistent with the ambiguity highlighted by Figure 6.

Table 10: counterfactual — increasing the weight on constituent exposure

Table 10: Counterfactual: making legislators more sensitive to the China shock (more negative δ)

	1997–2001	1993–1996	1990–1992
<i>Panel A: baseline</i>			
δ (range)	[−1.560, −0.075]	[−3.125, −0.125]	[−1.300, 0.000]
Pro-CHN vote share (%)	[57.663, 61.760]	[71.999, 77.034]	[37.682, 42.427]
<i>Panel B: set δ to the most negative value consistent with Democrats</i>			
δ (fixed)	−4.425	−6.325	−3.700
Pro-CHN vote share (%)	[17.485, 33.071]	[35.132, 58.356]	[11.983, 23.131]
<i>Panel C: set δ to the most negative value consistent with tight races</i>			
δ (fixed)	−3.413	−8.550	−4.375
Pro-CHN vote share (%)	[24.372, 42.458]	[24.606, 45.485]	[9.682, 19.424]

note: Panels B–C use extreme negative δ values taken from lower bounds of subgroup confidence sets (Democrats; and low win-margin legislators), then simulate vote shares.

Interpretation (Table 10).

- Unlike the “more information” counterfactual, changing *preferences/weights* matters a lot: making δ much more negative can drastically reduce pro-China votes.
- In panels B and C, simulated pro-China vote shares often fall well below 50% in 1997–2001 and 1993–1996, implying that if legislators had weighted constituency exposure as strongly as the most shock-sensitive subgroups, NTR renewal might not have passed.
- So the historical legislative outcomes look driven more by (i) ideology and baseline pro-trade preferences and (ii) a moderate (not extreme) weight on labor-market exposure, rather than by ignorance about the China shock.

9 A short summary

- Empirically, future China exposure is partially predictable from past exposure and manufacturing composition.
- The voting model implies $\Pr(Y = 1)$ depends on $\mathbb{E}[S_{i,t+1} \mid \mathcal{I}_{i,t}]$; different assumptions about what legislators know are testable using moment inequalities.
- Data reject a strong perfect-foresight hypothesis in most periods; baseline information is usually not rejected (especially for Democrats).
- Counterfactual perfect information barely changes vote outcomes; counterfactual stronger constituency sensitivity could overturn outcomes.

10 Conclusion

10.1 One sentence

To the title question “*Did US politicians expect the China shock?*” the takeaway is: **yes—politicians were fairly well informed about which districts would be hit harder—but**

they placed only a moderate weight on those expected local losses when voting on China NTR.

10.2 What is the paper doing?

The paper studies repeated U.S. House votes (1990–2001) on whether to renew China’s Normal Trade Relations (NTR) status. The key challenge is that the relevant economic consequences—the local labor-market impact of rising import competition from China—arrive *in the future*. So a legislator’s vote at time t depends on an *expectation* of a future district shock, $E[S_{i,t+1} | \mathcal{I}_{i,t}]$, and the econometrician does not observe $\mathcal{I}_{i,t}$ (the legislator’s information set).

The authors propose (i) a forward-looking voting model where the vote trades off ideology and expected electoral consequences from future trade shocks, and (ii) an estimation/testing strategy (moment inequalities) that lets them learn about both *preferences* (how much legislators care about constituents vs. ideology) and *information* (how much they could forecast about $S_{i,t+1}$) without requiring the econometrician to fully observe legislators’ information sets.

10.3 Core intuition: why expectations matter (the key identification problem)

A naive regression would plug in the *realized* future shock $S_{i,t+1}$ as if it were what legislators knew at the time of voting. But if legislators *did not* have perfect foresight, then using $S_{i,t+1}$ instead of $E[S_{i,t+1} | \mathcal{I}_{i,t}]$ is essentially a measurement-error problem. In a discrete choice model, this generates an *attenuation-type bias*: even if politicians truly respond to expected constituent losses (true $\delta < 0$), estimating on realized $S_{i,t+1}$ can make $\hat{\delta}$ look closer to zero. That creates a real interpretational risk:

a small estimated “constituent weight” could mean politicians are unresponsive (*preferences / accountability problem*) or simply that we falsely assumed they had more information than they actually did (*information problem*).

The paper’s main conceptual contribution is to build an empirical framework that separates these two margins.

10.4 Main findings

1. **Constituent exposure matters, but moderately.** Holding ideology fixed, districts with higher *expected* future China-shock exposure are less likely to have representatives voting pro-NTR (so $\delta < 0$). Quantitatively, an interquartile-range increase in expected exposure lowers the probability of a pro-China vote by only a few percentage points (roughly 3–5 pp in their discussion), whereas an analogous shift in ideology moves pro-China voting much more (about 13–17 pp). So ideology dominates, but constituency effects are not zero.
2. **Politicians had substantial information about the future shock.** Across years, the data are consistent with politicians knowing at least enough to forecast a large share of the cross-district variation in the future China shock (the paper emphasizes that they can typically forecast $\approx 55\%$ of the variation). In some early years they cannot reject near-perfect forecasting, but later they *do* reject perfect foresight.
3. **Information quality appears to deteriorate over the 1990s.** Predictability of the China shock declines over time (consistent with changing patterns of China’s comparative advantage and a weaker autocorrelation of industry-level shocks later). So “being informed” is not a fixed trait; it depends on how forecastable the underlying shock is.

4. **Heterogeneity: Democrats vs. Republicans, and electoral pressure.** Democrats look (i) more informed (baseline information not rejected more consistently) and (ii) more sensitive to constituent exposure (their δ is more clearly negative). Legislators facing tighter races put a heavier (more negative) weight on expected exposure than those in safe seats.
5. **Counterfactuals: more information changes little; more responsiveness changes a lot.** Giving everyone perfect information barely changes aggregate pro-NTR vote shares. In contrast, making everyone weight constituent exposure as strongly as the most shock-sensitive groups can flip outcomes against pro-China legislation.

10.5 So what is the conclusion, conceptually?

A synthesis of the conclusion is:

- The China-shock consequences were *not* a blind spot for legislators. They generally had reasonably accurate expectations about which districts would be exposed.
- The legislative outcomes were *not* primarily driven by ignorance/misinformation. If you give everyone perfect foresight, the model predicts very small changes in votes.
- Instead, the outcomes reflect a combination of (i) ideology/policy positioning and (ii) only moderate political weight placed on the groups that would later be harmed by import competition.
- Methodologically, the paper shows how moment-inequality tools can be used to study politicians' *information sets* (not just preferences) in a way that is robust to the econometrician not fully observing what decision-makers know.

10.6 Why this is useful

The paper gives us a clean way to think about policy decisions under uncertainty:

$$\textit{Observed policy choice} = \textbf{preferences} + \textbf{beliefs/information}.$$

If we do not explicitly model beliefs (and what information generated them), we can easily mis-diagnose the mechanism behind a weak correlation between policy votes and future realized consequences. This paper's approach is a template for disentangling whether a weak "policy response" is due to politicians not caring, or politicians not knowing.